TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

BE Programme | Year III / Part II

EMBEDDED SYSTEM (CT 655)

QUESTION BANK WITH ANSWERS

Chapter 8: IC Technology

COURSE SYLLABUS AT A GLANCE

Unit	Topic	Hours	Marks
1	Introduction to Embedded System	3	4
2	Hardware Design Issues	4	8
3	Software Design Issues	6	8
4	Memory	5	8
5	Interfacing	6	8
6	Real-Time Operating System (RTOS)	8	12
7	Control System	3	8
8	IC Technology	3	8
9	Microcontrollers in Embedded Systems	3	8
10	VHDL	4	8
	Total	**45**	**80**

Chapter 8 Topics (3 Hours | 8 Marks):

- 8.1 Full-Custom (VLSI) IC Technology — Design Steps, Advantages, Disadvantages
- 8.2 IC Manufacturing Process — Silicon Wafer to Packaged Chip
- 8.3 Photolithography — Process Steps, Types (Contact, Projection, EUV, E-beam)
- 8.4 Semi-Custom (ASIC) IC Technology — Gate Array, Standard Cell
- 8.5 Programmable Logic Device (PLD) IC Technology — SPLD, CPLD, FPGA
- 8.6 Comparison: Full-Custom vs Semi-Custom vs PLD
- 8.7 CMOS Technology — NAND, NOR Gate Design using CMOS

Chapter 8: IC Technology

Q1. What are the steps involved in manufacturing of an IC? Show the various steps of the photolithography process using appropriate diagrams.	**2082 Kartik (Back) | 2081 Ashwin (Back) | 2080 Ashwin (Back) | 2075 Bhadra (Back) [3+5]**

ANSWER

IC Manufacturing — Overview:

```
Silicon Sand → [Purification] → [Czochralski Growth] → Silicon Ingot
→ [Wafer Slicing] → [Polishing] → Silicon Wafer
→ [Epitaxial Growth] → [Repeated: Photolithography + Etch + Dope +
Deposit]
→ [Metallization] → [CMP] → [Wafer Test] → [Dicing] → [Packaging] → IC
```

Detailed IC Manufacturing Steps:

1. Silicon Ingot Growth (Czochralski Process): Ultra-pure polycrystalline silicon is melted in a quartz crucible. A seed crystal is dipped in and slowly pulled while rotating, growing a single-crystal cylindrical silicon ingot (diameter 200mm or 300mm).
2. Wafer Slicing and Polishing: The ingot is sliced into thin wafers (~0.5–1 mm thick) using a diamond wire saw. Wafers are lapped and chemically mechanically polished (CMP) to a mirror-smooth surface (roughness < 0.1 nm).
3. Epitaxial Layer Growth: A thin (1–10 μm), precisely doped silicon layer is grown on the wafer surface by Chemical Vapor Deposition (CVD). This provides a clean, controlled starting layer for transistor fabrication.
4. Photolithography: Pattern transfer from photomask to wafer surface using UV/EUV light and photoresist. Repeated for every layer (~20–30 cycles in modern chips). (See detailed steps below.)
5. Etching: Remove exposed material to create circuit features. Wet etching uses chemical solutions (HF for SiO_2). Dry/Plasma etching uses ionized gas (more precise, anisotropic — used for sub-micron features).
6. Doping (Ion Implantation): Dopant ions (Boron for p-type, Phosphorus/Arsenic for n-type) are accelerated and implanted into exposed silicon regions to form source, drain, and well regions of MOSFETs.
7. Thin Film Deposition: Insulating layers (SiO_2, Si_3N_4) by CVD; conducting layers (polysilicon for gates, tungsten for vias, copper for interconnects) by CVD or PVD (sputtering).
8. Chemical Mechanical Planarization (CMP): Polishes each deposited layer flat to allow accurate multi-level metallization (modern chips have 10–15 metal layers).
9. Metallization: Deposit and pattern metal interconnect layers (aluminum in older processes, copper in modern) to connect all transistors and form circuit wiring.

10. Wafer Testing (Probe Test): Automated probe stations electrically test every die on the wafer. Failing dies are inked/marked.
11. Dicing: Diamond saw or laser dices the wafer into individual chips (dies).
12. Packaging: Die is bonded to a lead frame or substrate; bond wires connect die pads to package pins; encapsulated in plastic (DIP, QFP, BGA) or ceramic.
13. Final Testing: Packaged IC tested at full speed, temperature extremes, and burn-in for reliability screening.

Photolithography Process Steps (one layer cycle):

```
Step 1: OXIDATION
  Wafer heated to 900-1200°C in O2/H2O atmosphere
  → SiO2 grows on silicon surface (~100nm thick)

  ┌──────────────────────────────┐
  │   SiO2 (oxide layer)         │  ← grown oxide
  │   Silicon Wafer              │
  └──────────────────────────────┘


Step 2: PHOTORESIST APPLICATION (Spin Coating)
  Liquid photoresist spun onto wafer at high RPM → uniform thin film

  ┌──────────────────────────────┐
  │▒▒▒ Photoresist (PR) ▒▒▒▒▒▒▒▒▒│  ← ~1µm thick
  │   SiO2                       │
  │   Silicon Wafer              │
  └──────────────────────────────┘


Step 3: SOFT BAKE  →  ~90°C, evaporates solvent, hardens PR

Step 4: MASK ALIGNMENT + UV EXPOSURE
  Photomask (quartz + chrome pattern) aligned over wafer
  UV light exposes through clear areas of mask

  ┌──────────────────────────────┐
  │▒▒ Chrome ▒▒│  Clear  │▒▒▒▒▒▒│  ← Photomask
  │──────────────────────────────│  UV light ↓
  │▒▒▒▒▒▒▒▒▒▒▒▒▒│ exposed │▒▒▒▒▒▒│  ← PR (exposed = soluble)
  │   SiO2                       │
  └──────────────────────────────┘


Step 5: DEVELOPMENT
  Developer dissolves exposed positive resist → pattern revealed

  ┌──────────────────────────────┐
  │▒▒▒▒▒▒▒▒▒▒▒▒▒│         │▒▒▒▒▒▒│  ← remaining PR (hardened)
  │   SiO2      │  open   │      │  ← oxide exposed
  └──────────────────────────────┘


Step 6: HARD BAKE  →  ~120°C, further hardens remaining resist

Step 7: ETCHING
  HF acid etches exposed SiO2; resist protects other areas
```

```
   ┌──────────────────────────────┐
   │▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ |          |▒▒▒▒▒▒▒ |  ← resist still present
   |    SiO2      |  window |        |  ← oxide removed
   |    Silicon   |▓▓▓▓▓▓▓▓▓|        |  ← silicon exposed
   └──────────────────────────────┘


 Step 8: RESIST STRIPPING
   Remaining photoresist removed with solvent/plasma
   ┌──────────────────────────────┐
   |    SiO2      |  window |        |  ← final patterned oxide
   |    Silicon Wafer               |
   └──────────────────────────────┘


   → Repeat all 8 steps for each subsequent layer (doping, gate, metal...)
```

Q2. Discuss the advantages and disadvantages of full-custom IC technology. What types of products use full-custom ICs?	**2072 Ashwin (Regular) | 2078 Chaitra (Back) | 2074 Jestha (Back) [3+5]**

ANSWER

Full-Custom IC Technology:

In full-custom (VLSI) design, every transistor, every wire, and every cell is designed manually at the mask layout level by experienced VLSI designers. Nothing is pre-placed or pre-characterized — every dimension is optimized for the specific application.

Advantages	Disadvantages
Maximum performance: Transistors sized and placed for minimum delay; critical paths fully optimized	Very high NRE (Non-Recurring Engineering) cost: Full mask set for advanced node (7nm) costs $20M–$100M+
Minimum chip area: No wasted silicon — every transistor is optimally placed and sized	Extremely long design time: Months to years for complex chips; large expert team required
Minimum power consumption: Transistor sizing minimizes switching power	Requires highly skilled VLSI designers: Rare and expensive expertise needed
Best for analog and mixed-signal circuits: Precise matching, custom layout critical for analog performance	High risk: A single mask error requires expensive re-fabrication (re-spin)
Competitive advantage: Unique performance impossible with semi-custom	Only economical at very high volume: NRE amortized only over millions of units
Full control over every design decision: No library constraints	Longer time-to-market compared to FPGA/standard cell

Products Using Full-Custom ICs:

- Microprocessors and GPUs: Intel Core, AMD Ryzen, NVIDIA A100 — critical datapaths hand-optimized.

- Mobile SoCs: Apple A-series, Qualcomm Snapdragon — billions of units justify NRE.
- Memory ICs: DRAM and NAND Flash cells are full-custom for maximum density (Samsung, Micron).
- Analog ICs: Precision ADCs, PLLs, RF circuits — layout matching and parasitics critical.
- High-speed SerDes: 100 Gbps+ transceivers — only custom design meets timing.

Q3. What are the merits and demerits of full-custom and semi-custom IC technology? Explain gate-array and standard-cell semi-custom IC technologies.	**2081 Chaitra (Regular) | 2079 Chaitra (Back) | 2073 Bhadra (Back) [4+4]**

ANSWER

Feature	Full-Custom	Semi-Custom (ASIC)
Performance	Highest — every transistor optimized	Good — slightly less than full-custom
Design Time	Very long (months–years)	Moderate (weeks–months)
NRE Cost	Very high (full mask set, millions)	Moderate (partial or full masks, $500K–$5M)
Flexibility	Maximum optimization freedom	Limited to library cells or pre-placed transistors
Chip Area	Minimum	10–30% larger than full-custom
Designer skill	Expert VLSI layout designers	RTL/synthesis engineers (more common)
Volume needed	Very high (millions+)	Medium to high (thousands to millions)
Examples	Intel CPU, Apple A-chip	Telecom ASICs, networking chips

Gate-Array Semi-Custom IC Technology:

A gate-array IC is prefabricated with a regular array of uncommitted transistor cells on a silicon wafer up to — but not including — the metal interconnect layers. Wafers are stockpiled and only the metal mask layers are customized per design.

```
Gate-Array Structure:
 ┌──────────────────────────────────────────────┐
 │  I/O Pads                                    │
 │  ┌──────────────────────────────────────────┐ │
 │  │ Row: [Cell][Cell][Cell]...[Cell]         │ │
 │  │      ← Routing Channel →                 │ │
 │  │ Row: [Cell][Cell][Cell]...[Cell]         │ │
 │  │      ← Routing Channel →                 │ │
 │  │ Row: [Cell][Cell][Cell]...[Cell]         │ │
 │  └──────────────────────────────────────────┘ │
 └──────────────────────────────────────────────┘
   All cells pre-fabricated (transistors placed)
```

```
  Only METAL layers are custom — fast turnaround
```

- Advantage: Short time-to-market (days to weeks once design is complete); lower cost than full-custom since most fabrication steps are shared (pre-fabricated wafer stockpile).
- Disadvantage: Transistors are pre-placed in fixed positions — cannot optimize cell sizes. Some cells may not be used (wasted area). Routing channels reduce density.
- Types: Channeled gate array (routing between rows), channel-less/Sea-of-Gates (denser, routing over cells).

Standard Cell Semi-Custom IC Technology:

Standard cells are pre-designed, pre-characterized logic cells (NAND2, NOR2, DFF, adder, MUX) with fixed height but variable width, stored in a technology library.

```
Standard Cell Design Flow:
  HDL (VHDL/Verilog) → [Synthesis] → Netlist (gates)
       → [Floorplanning] → [Placement] → [Clock Tree Synthesis]
       → [Routing] → [STA] → [DRC/LVS] → GDSII → Fabrication

Standard Cell Row Layout:
┌──────────────────────────────────────────────┐
│ NAND │ INV │ DFF  │ NOR │ MUX │ BUF │ AND    │  ← row of cells
├──────────────────────────────────────────────┤  VDD/GND rails
│ DFF  │ XOR │ NAND │ INV │ DFF │ OR  │ NAND   │  ← next row
└──────────────────────────────────────────────┘
  Variable width, fixed height — rows abutted, VDD/GND shared
```

- Advantage: Better area and performance than gate arrays (cells individually optimized). EDA tools (Synopsys, Cadence) automate synthesis, placement, and routing. Most widely used ASIC methodology today.
- Disadvantage: Full mask set required — higher NRE than gate arrays. Limited to available cell library functions.
- Used by: Qualcomm, MediaTek, ARM, Broadcom — most commercial ASICs.

Q4. Describe PLD (Programmable Logic Device) IC technology. Explain SPLD, CPLD, and FPGA with diagrams.	**2078 Poush (Back) | 2073 Bhadra (Back) | 2076 Chaitra (Back) [8]**

ANSWER

PLD (Programmable Logic Device) Overview:

PLDs are ICs that can be configured by the designer after manufacture to implement any desired digital logic function — no custom fabrication required. They are programmed using a hardware description language (VHDL/Verilog) and a software toolchain.

1. SPLD — Simple PLD (PAL/GAL):

SPLDs contain a programmable AND array and a fixed OR array, or both programmable AND and OR arrays. They implement sum-of-products (SOP) Boolean expressions.

```
SPLD (PAL) Internal Structure:

Inputs  ──►┌──────────────────────────────┐
 A ────────►│  Programmable AND Array      │──► Product Terms
 B ────────►│  (fuse/antifuse matrix)      │──► (AND gates)
 C ────────►│                              │──► P1 = A·B'
            └──────────────────────────────┘    P2 = A'·C
                         │                      P3 = B·C
                         ▼
            ┌──────────────────────────────┐
            │  Fixed OR Array              │──► F1 = P1+P2
            │                              │──► F2 = P2+P3
            └──────────────────────────────┘
                         │
            ┌────────────┴─────────────────┐
            │  Output Logic (optional FF)  │──► Outputs
            └──────────────────────────────┘

Programming: Blow fuses in AND array to select inputs for each product
term
Capacity: Tens to hundreds of gates
Examples: PAL16L8, GAL22V10
```

2. CPLD — Complex PLD:

A CPLD consists of multiple SPLD-like macrocell blocks connected by a programmable interconnect matrix (PIM). Offers much higher capacity than SPLDs while maintaining predictable timing.

```
CPLD Structure:
┌──────────────────────────────────────────────────────┐
│                        CPLD                          │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐           │
│  │ Macro-   │  │ Macro-   │  │ Macro-   │           │
│  │ cell     │  │ cell     │  │ cell     │           │
│  │ Block 1  │  │ Block 2  │  │ Block 3  │           │
│  │(AND-OR   │  │(AND-OR   │  │(AND-OR   │           │
│  │ array)   │  │ array)   │  │ array)   │           │
│  └────┬─────┘  └────┬─────┘  └────┬─────┘           │
│       └──────────────┴──────────────┘                │
│           Programmable Interconnect Matrix (PIM)     │
│                 (switch matrix)                      │
│  I/O Blocks ◄──────────────────────────────────────  │
└──────────────────────────────────────────────────────┘

Capacity: Hundreds to thousands of gates
Programming: EEPROM or Flash cells (non-volatile, retains config at power-
off)
Timing: Predictable (pin-to-pin delay specified, unlike FPGA)
```

```
Examples: Altera MAX series, Xilinx XC9500, Lattice ispMACH
```

3. FPGA — Field Programmable Gate Array:

FPGAs are the most powerful and flexible PLDs. They consist of a large array of configurable logic blocks (CLBs), connected by a programmable routing network, with I/O blocks around the perimeter. Modern FPGAs also include hard IP blocks (multipliers, RAM, DSP slices, PCIe, Ethernet MACs, ARM cores).

```
FPGA Structure:
┌──────────────────────────────────────────────────────────────┐
│  I/O   │                    FPGA                   │  I/O   │
│ Blocks │  ┌─────┐   ┌─────┐   ┌─────┐   ┌─────┐     │ Blocks │
│        │  │ CLB │──│ CLB │──│ CLB │──│ CLB │     │        │
│        │  └─────┘   └─────┘   └─────┘   └─────┘     │        │
│        │     │  Prog. │  Route │  Matrix│           │        │
│        │  ┌─────┐   ┌─────┐   ┌─────┐   ┌─────┐     │        │
│        │  │ CLB │   │ CLB │   │ DSP │   │BRAM │     │        │
│        │  └─────┘   └─────┘   └─────┘   └─────┘     │        │
│        │    Programmable Routing Network            │        │
└──────────────────────────────────────────────────────────────┘


CLB (Configurable Logic Block) contains:
  • LUT (Look-Up Table): 4-6 input truth table → implements any Boolean fn
  • Flip-Flop (D-FF): Sequential logic / registers
  • MUX: Connects LUT output to FF or directly to routing

Configuration: SRAM-based (most common — reprogrammable, volatile — needs
  bitstream reload at power-on) or Flash/Antifuse (non-volatile)
Capacity: Thousands to millions of LUTs
Examples: Xilinx Artix-7, Virtex UltraScale+; Intel Cyclone, Stratix
```

SPLD vs CPLD vs FPGA Comparison:

Property	SPLD (PAL/GAL)	CPLD	FPGA
Capacity	Small (tens of gates)	Medium (hundreds–thousands)	Large (thousands–millions of LUTs)
Architecture	AND-OR array	Multiple AND-OR macrocell blocks + PIM	LUT array + programmable routing + hard IP
Configuration	Fuse/EEPROM	EEPROM/Flash (non-volatile)	SRAM (volatile) or Flash/Antifuse
Timing	Predictable	Predictable pin-to-pin delay	Less predictable (routing dependent)
Power	Low	Low-Medium	Medium-High (many SRAM bits)

Non-volatile	Yes (most)	Yes	No (SRAM) / Yes (Flash/Antifuse)
Reprogrammable	Yes (GAL)	Yes	Yes (SRAM, many times)
Examples	PAL16L8, GAL22V10	MAX7000, XC9500	Artix-7, Cyclone V, Virtex-7
Best used for	Simple glue logic	Interface, control logic	Complex DSP, prototyping, large designs

Q5. Describe the design steps involved for Full-Custom, Semi-Custom, and PLD IC technologies.	**2078 Poush (Back) | 2074 Bhadra (Back) | 2076 Baisakh (Back) [8]**

ANSWER

Full-Custom IC Design Steps:

```
1. System Specification → 2. Architecture Design → 3. Circuit Design
     → 4. Manual Layout → 5. DRC → 6. LVS → 7. Parasitic Extraction
     → 8. Post-Layout Simulation → 9. GDSII Generation → 10. Fabrication
```

1. System Specification: Define functional requirements, performance targets (speed, power), area budget, and process technology node.
2. Architecture Design: Partition into functional blocks (datapath, control, memory, I/O). Define interface signals.
3. Circuit Design: Design transistor-level schematics. Size each MOSFET (W/L ratio) for required drive strength, speed, and power. Simulate using SPICE.
4. Manual Layout: Draw every mask layer (active, poly, n-well, contacts, metal1, metal2...) using Virtuoso/Laker. Every transistor is manually placed and sized.
5. Design Rule Check (DRC): Verify layout satisfies fabrication constraints (minimum width, spacing, overlap rules) for the chosen process node.
6. Layout vs. Schematic (LVS): Extract netlist from layout and compare with original schematic — must match exactly.
7. Parasitic Extraction and Post-Layout Simulation: Extract resistances and capacitances from layout; re-simulate timing to verify performance with actual parasitics.
8. GDSII Generation: Generate GDSII (Graphic Database System II) file — the standard format for photomask fabrication.
9. Fabrication, Packaging, and Testing: (See Q1.)

Semi-Custom (Standard Cell ASIC) Design Steps:

```
RTL (VHDL/Verilog) → [Synthesis] → Gate Netlist
  → [Floorplan] → [Placement] → [CTS] → [Routing]
  → [STA] → [DRC/LVS] → [GDSII] → Fabrication
```

1. RTL Design (HDL Coding): Write Register-Transfer Level description in VHDL or Verilog. Simulate with testbench for functional verification.

2. Logic Synthesis: Synthesis tool (Synopsys Design Compiler) maps RTL to standard cells from the foundry's cell library. Optimizes for timing, area, or power.
3. Gate-Level Simulation: Verify synthesized netlist against RTL simulation results. Check for synthesis issues.
4. Floorplanning: Define chip boundary; plan placement of large blocks (memories, analog blocks); place I/O pads; define power grid (VDD/GND rings and stripes).
5. Placement: Place all standard cells in rows within floorplan. Tools minimize wire lengths to meet timing constraints.
6. Clock Tree Synthesis (CTS): Build a balanced clock distribution network to minimize clock skew between all flip-flops.
7. Routing: Connect all cell pins with metal wires following design rules. Fill remaining area with filler cells.
8. Static Timing Analysis (STA): Verify all setup and hold timing constraints are met with actual wire delays (using Synopsys PrimeTime).
9. DRC/LVS, GDSII generation, Fabrication, and Testing.

PLD (FPGA) Design Steps:

```
HDL/Schematic → [Synthesis] → [Implementation: Place&Route]
  → [Timing Analysis] → [Bitstream Generation] → [JTAG Programming] → Test
```

1. Design Entry: Write HDL (VHDL/Verilog) or use schematic capture / IP block integration.
2. Functional Simulation: Verify RTL behavior with testbench before synthesis.
3. Synthesis: Xilinx Vivado / Intel Quartus synthesizes HDL into an FPGA netlist (LUTs, FFs, BRAMs, DSPs).
4. Implementation (Place and Route): Tool places LUTs/FFs onto specific FPGA sites and routes connections through the programmable routing fabric.
5. Timing Analysis: Verify all timing constraints (setup/hold, clock period) are met with actual FPGA routing delays.
6. Bitstream Generation: Create device-specific binary configuration file (.bit for Xilinx, .sof for Intel).
7. Device Programming: Download bitstream to FPGA via JTAG cable (Xilinx Platform Cable, Intel USB-Blaster). Configuration stored in SRAM (needs reload at power-on) or external SPI Flash (auto-loads).
8. Hardware Testing: Verify design on target board using logic analyzer, oscilloscope, or embedded logic analyzer (ILA/SignalTap).

Q6. Define photolithography and explain its types in the context of IC manufacturing.	**2080 Ashwin (Back) | 2079 Ashwin (Back) | 2077 Chaitra (Back) [4+4]**

ANSWER

Photolithography:

Photolithography (also called optical lithography or microlithography) is the process of transferring geometric patterns from a photomask to the surface of a semiconductor wafer using light exposure and light-sensitive photoresist chemistry. It is the most critical and repeated step in IC fabrication — performed 20–30 times per chip to define each layer.

Types of Photolithography:

1. Contact Lithography: The photomask is placed in direct physical contact with the photoresist-coated wafer during exposure.

- Resolution: ~1–2 μm (limited by diffraction and mask wear).
- Advantage: Simple, inexpensive equipment.
- Disadvantage: Physical contact damages mask quickly; defects from dust particles; poor yield. Used only in research and low-volume old-process fabrication.

2. Proximity Lithography: A small gap (10–50 μm) is maintained between mask and wafer during exposure.

- Resolution: ~2–5 μm (worse than contact due to diffraction across gap).
- Advantage: Eliminates mask wear problem compared to contact lithography.
- Disadvantage: Lower resolution due to diffraction across the gap. Not suitable for modern IC features.

3. Projection Lithography (Most Important): A reduction optical system projects the mask image (typically 4× or 5× reduced) onto the wafer. Used in all modern IC manufacturing.

- Resolution: Sub-100nm with advanced techniques (DUV, immersion, multiple patterning).
- Sub-types: Stepper (exposes one die at a time, then steps to next position) and Scanner (slit scans across die — better uniformity, used in modern fabs).
- Light sources: i-line (365 nm), KrF (248 nm), ArF (193 nm DUV — most common today), ArF immersion (193 nm with water immersion — achieves ~40 nm features).
- Advantage: High resolution, mask reused thousands of times, mass production.

4. Electron Beam (E-Beam) Lithography: A focused electron beam directly writes patterns onto the resist without a mask.

- Resolution: Sub-10 nm (highest resolution available, ~1–5 nm in research).
- Advantage: No mask needed — direct write; used for mask fabrication itself and research.
- Disadvantage: Extremely slow (serial write, not parallel) — cannot be used for high-volume wafer production.

5. EUV (Extreme Ultraviolet) Lithography: Uses 13.5 nm wavelength light generated by a high-energy laser hitting a tin droplet target. All optics are reflective mirrors (no lenses work at EUV wavelengths).

- Resolution: Sub-5 nm features (currently used for 3nm, 2nm nodes by TSMC and Samsung).
- Advantage: Enables continued Moore's Law scaling; eliminates multiple patterning complexity of DUV.

- Disadvantage: Extremely expensive equipment ($150M+ per scanner); requires near-perfect vacuum; low light source power limits throughput.

Type	Wavelength	Resolution	Volume Production	Use Case
Contact	UV (365nm)	~1–2 µm	No (mask wear)	Research, old processes
Proximity	UV (365nm)	~2–5 µm	No	Research only
Projection (DUV)	193nm (ArF)	40–130nm	Yes — mainstream	14nm–130nm nodes
Immersion DUV	193nm + H2O	~40nm (single)	Yes — mainstream	7nm–40nm (with multi-patterning)
E-Beam	Electrons (~1nm)	<10nm	No (too slow)	Mask writing, research
EUV	13.5nm	<5nm	Yes — cutting edge	3nm, 2nm, 1nm nodes

Q7. Compare Full-Custom, Semi-Custom (ASIC), and PLD (FPGA) IC technologies from an embedded system designer's perspective.	**2081 Chaitra (Regular) | 2080 Chaitra (Back) | 2074 Chaitra (Back) [8]**

ANSWER

Property	Full-Custom (VLSI)	Semi-Custom (ASIC)	PLD (FPGA/CPLD)
Performance	Highest — manually optimized	High — synthesis-optimized cells	Moderate — routing overhead
Power	Lowest — optimized transistors	Low-moderate	Higher (SRAM config overhead)
Chip Area	Smallest	Moderate (10–30% more than full-custom)	Largest (LUT overhead)
NRE Cost	Very high ($5M–$100M+)	Moderate ($500K–$5M)	Zero (off-shelf chip)
Unit Cost	Lowest (at high volume)	Low (at medium-high volume)	Higher per unit
Design Time	Months to years	Weeks to months	Days to weeks
Time-to-Market	Longest	Moderate	Fastest
Flexibility	None after tapeout	None after tapeout	Fully reprogrammable in field
Volume Economics	Best at millions+ units	Best at thousands–millions units	Best at small–medium volumes
Tools	Virtuoso, Calibre (expensive)	Synopsys, Cadence (expensive)	Vivado, Quartus (free/low cost)
Non-volatile	N/A	N/A	SRAM=No; Flash/Antifuse=Yes

Examples	CPU, GPU, DRAM	Networking chips, baseband modems	Prototyping, industrial ctrl, video processing

Selection Guidelines for Embedded System Designers:

Choose Full-Custom when: Product volume is extremely high (10M+ units/year), performance is critical, power budget is very tight, and development budget is large (smartphones, high-end processors, memory).

Choose Semi-Custom (ASIC) when: Volume justifies NRE (100K+ units), design is finalized and won't change, better performance/power than FPGA needed (telecom, automotive, IoT at scale).

Choose FPGA when: Design needs flexibility (field updates, algorithm changes), time-to-market is critical, volume is low-to-medium, or for prototyping before ASIC tapeout. Also when design is too large for CPLD but ASIC NRE not justified.

Choose CPLD when: Simple glue logic, interface bridging, power-on configuration logic — small designs needing non-volatile, fast, predictable timing.

Q8. Explain CMOS technology. Design CMOS NAND gate and CMOS NOR gate with circuit diagrams and truth tables.	**2079 Ashwin (Back) | 2078 Baishakh (Back) | 2076 Baisakh (Back) [3+5]**

ANSWER

CMOS (Complementary Metal-Oxide-Semiconductor) Technology:

CMOS is the dominant IC fabrication technology. It uses complementary pairs of NMOS (n-channel) and PMOS (p-channel) MOSFETs. The key property is that at steady state, one transistor is always OFF — no direct DC path from VDD to GND — giving near-zero static power consumption.

- NMOS transistor: ON when gate = HIGH (VGS > Vth). Connects output to GND (pull-down network, PDN).
- PMOS transistor: ON when gate = LOW (VGS < −|Vth|). Connects output to VDD (pull-up network, PUN).
- Static power: Near zero (only leakage current when both transistors off simultaneously).
- Dynamic power: $P = \alpha \cdot C \cdot VDD^2 \cdot f$ — proportional to switching frequency, load capacitance, and VDD^2.

CMOS NAND Gate (2-input):

```
                VDD
                 │
            ┌────┴────┐
A ──────────┤ PMOS P1 │  (A=0 → P1 ON)
            └────┬────┘
                 │
            ┌────┴────┐
```

```
B ──────┤ PMOS P2 │  (B=0 → P2 ON)
        └─────────┘
             │
             ├──────────── Y = (A·B)' output
             │
        ┌────────┐
A ──────┤ NMOS N1 │  (A=1 → N1 ON)
        └────────┘
             │
        ┌────────┐
B ──────┤ NMOS N2 │  (B=1 → N2 ON)
        └────────┘
             │
            GND

PUN: P1 and P2 in PARALLEL — output HIGH if A=0 OR B=0
PDN: N1 and N2 in SERIES  — output LOW  if A=1 AND B=1
```

A	B	N1 (NMOS)	N2 (NMOS)	P1 (PMOS)	P2 (PMOS)	Y = (AB)'
0	0	OFF	OFF	ON	ON	1 (VDD through P1 and P2)
0	1	OFF	ON	ON	OFF	1 (VDD through P1)
1	0	ON	OFF	OFF	ON	1 (VDD through P2)
1	1	ON	ON	OFF	OFF	0 (GND through N1 and N2)

CMOS NOR Gate (2-input):

```
            VDD
             │
        ┌────────┐
A ──────┤ PMOS P1 │  (A=0 → P1 ON)
        └────────┘
             │
        ┌────────┐
B ──────┤ PMOS P2 │  (B=0 → P2 ON)
        └────────┘
             │
             ├──────────── Y = (A+B)' output
        ┌────┴───┐   ┌────┴───┐
A ──────┤ NMOS N1 │   │ NMOS N2 ├── B
        └────────┘   └────────┘
             └──────┬──────┘
                    GND

PUN: P1 and P2 in SERIES   — output HIGH only if A=0 AND B=0
PDN: N1 and N2 in PARALLEL — output LOW  if A=1 OR  B=1
```

A	B	N1 (NMOS)	N2 (NMOS)	P1 (PMOS)	P2 (PMOS)	Y = (A+B)'
0	0	OFF	OFF	ON	ON	1 (VDD through P1 and P2 in series)
0	1	OFF	ON	ON	OFF	0 (GND through N2)
1	0	ON	OFF	OFF	ON	0 (GND through N1)
1	1	ON	ON	OFF	OFF	0 (GND through N1 or N2)

Key Design Rule for CMOS Gates:

- PUN (Pull-Up Network) is the dual of PDN (Pull-Down Network): Series in PDN → Parallel in PUN; Parallel in PDN → Series in PUN.
- NAND gate: PDN = series NMOS (both must be ON to pull down) → PUN = parallel PMOS (either can pull up).
- NOR gate: PDN = parallel NMOS (either pulls down) → PUN = series PMOS (both must be ON to pull up).
- PMOS transistors must be wider (typically 2–3× W) than NMOS for equal drive strength due to lower hole mobility.

Q9. Draw and explain the CMOS inverter circuit. What is propagation delay and how does it affect IC performance?	**Model Question | 2075 Chaitra (Back) [4+4]**

ANSWER

CMOS Inverter:

```
            VDD
             |
        ┌────┴────┐
A ──────┤  PMOS   │   (when A=0: PMOS ON, NMOS OFF → Y=VDD=1)
        └────┬────┘
             |
             |
             ├──────── Y = A'   (output)
             |
        ┌────┴────┐
A ──────┤  NMOS   │   (when A=1: NMOS ON, PMOS OFF → Y=GND=0)
        └────┬────┘
             |
            GND

Truth table: A=0 → Y=1;  A=1 → Y=0   (NOT gate)
Static power: ~0 (one transistor always OFF — no VDD-to-GND path)
Dynamic power: P = C·VDD²·f  (charges/discharges load capacitance C)
```

Operation:

- Input A = LOW (0V): PMOS gate-source voltage VGS = −VDD < −|Vthp| → PMOS ON. NMOS VGS = 0 < Vthn → NMOS OFF. Output Y charged to VDD = logic HIGH.

- Input A = HIGH (VDD): NMOS VGS = VDD > Vthn → NMOS ON. PMOS VGS = 0 > −|Vthp| → PMOS OFF. Output Y discharged to GND = logic LOW.
- At any stable state: One transistor ON, one OFF — no static DC current path. Near-zero static power.

Propagation Delay:

Propagation delay (tp) is the time from when the input transitions 50% to when the output transitions 50%. It is the fundamental measure of logic gate speed.

Parameter	Symbol	Definition
Rise time	tr	Time for output to transition from 10% to 90% of VDD
Fall time	tf	Time for output to fall from 90% to 10% of VDD
High-to-Low delay	tpHL	Input rises 50% → output falls to 50%
Low-to-High delay	tpLH	Input falls 50% → output rises to 50%
Propagation delay	tp	Average: tp = (tpHL + tpLH) / 2

Factors affecting propagation delay:

- Load capacitance (CL): tp □ CL — larger fan-out or wire capacitance → slower. tp = CL × VDD / (2 × Ion).
- Supply voltage (VDD): Lower VDD → less drive current → slower switching. tp ∝ 1/VDD.
- Transistor size (W/L): Wider transistors → more current → faster. But larger area → more capacitance.
- Temperature: Higher temperature → lower carrier mobility → slower switching.
- Process node: Smaller transistors (7nm vs 28nm) → shorter channels → faster switching, lower capacitance.
- Impact on IC performance: Maximum clock frequency fmax ≈ 1/(tp × N_stages) where N_stages is pipeline depth. Minimizing propagation delay is the primary goal of VLSI optimization.

Summary: Topics Covered in Chapter 8

S.N.	Topic	Exam(s) Asked	Marks
1	IC Manufacturing Steps (13 steps, full process)	2082 Kartik, 2081 Ashwin, 2080 Ashwin, 2075 Bhadra (annual)	3–8
2	Photolithography Process Steps (8 steps with ASCII diagrams)	Almost every exam — most important topic	5–8
3	Types of Photolithography (Contact/Proximity/Projection/E-Beam/EUV)	2080 Ashwin, 2079 Ashwin, 2077 Chaitra	4+4
4	Full-Custom IC — Advantages and Disadvantages	2072 Ashwin, 2078 Chaitra, 2074 Jestha	3+5
5	Semi-Custom IC — Full-Custom vs Semi-Custom comparison table	2081 Chaitra, 2079 Chaitra, 2073 Bhadra	4+4
6	Gate-Array Semi-Custom — structure + advantages/disadvantages	2081 Chaitra, 2079 Chaitra, 2073 Bhadra	4
7	Standard Cell Semi-Custom — design flow + advantages	2081 Chaitra, 2079 Chaitra, 2073 Bhadra	4
8	PLD — SPLD/PAL structure with diagram	2078 Poush, 2073 Bhadra, 2076 Chaitra	4
9	CPLD — structure, macrocell blocks, PIM diagram	2078 Poush, 2073 Bhadra, 2076 Chaitra	4
10	FPGA — CLB/LUT structure, SRAM configuration, hard IP	2078 Poush, 2073 Bhadra, 2076 Chaitra	4
11	SPLD vs CPLD vs FPGA full comparison table	2078 Poush, 2073 Bhadra	4+4
12	Design Steps: Full-Custom (9 steps)	2078 Poush, 2074 Bhadra, 2076 Baisakh	8
13	Design Steps: Semi-Custom ASIC (10 steps)	2078 Poush, 2074 Bhadra	8
14	Design Steps: PLD/FPGA (8 steps)	2078 Poush, 2073 Bhadra	8
15	Full-Custom vs Semi-Custom vs PLD — full comparison table	2081 Chaitra, 2080 Chaitra, 2074 Chaitra	8
16	CMOS Technology + NAND gate (circuit + truth table)	2079 Ashwin, 2078 Baishakh, 2076 Baisakh (annual)	3+5
17	CMOS NOR gate (circuit + truth table)	2079 Ashwin, 2078 Baishakh, 2076 Baisakh	3+5
18	CMOS Inverter + Propagation Delay	2075 Chaitra, Model Q	4+4